

Mountain goat surveys on Kodiak Island are a cooperation between ADFG and USFWS. Mountain goats were introduced in 1952 (18 animals) and now number over 3,000. There is a major concern of habitat degradation caused by overpopulation of goats on the island, and the effect that could have on native species such as bears. Maintaining the population for hunter opportunities is also a goal, however.

There are 8 hunt areas on the island. USFWS is responsible for surveying hunt area 480 and ADFG is responsible for all other areas.

Due to the growth of the goat population in HA 480, ADFG has liberalized hunting in the area to curb population growth; in 2010 the area was made a registration hunt, in xx the bag limit was increased to two animals, and in 2019 the season was lengthened and one of the two animals killed was required to be a female.

Methods

There are 25 survey units within the survey area.

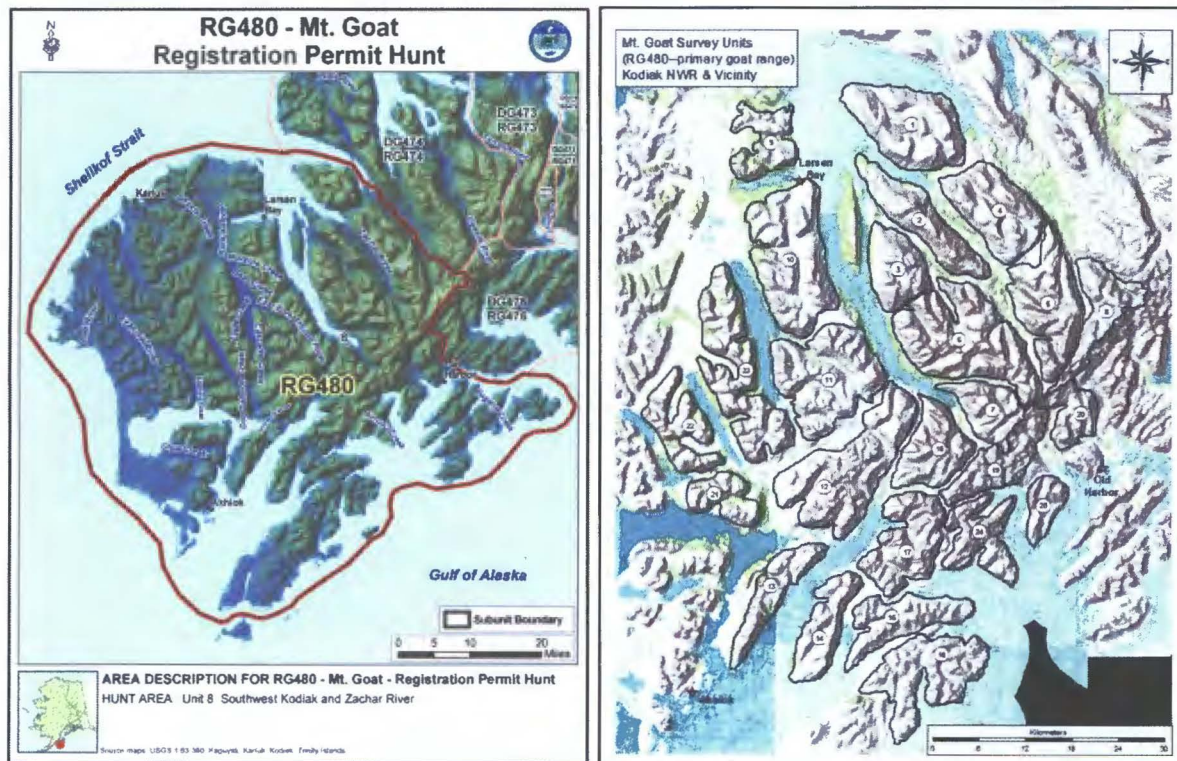
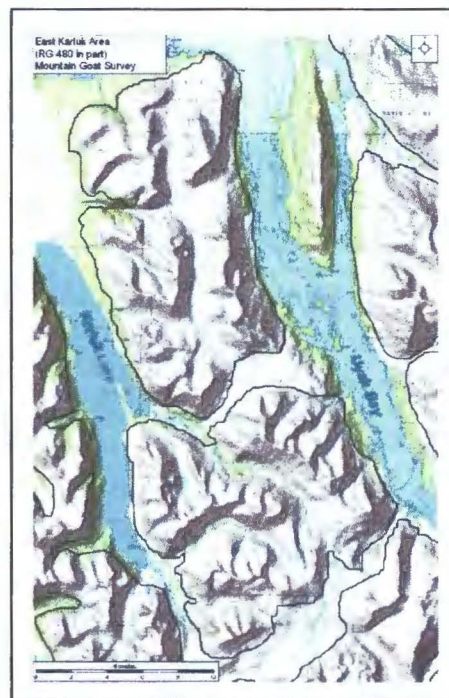
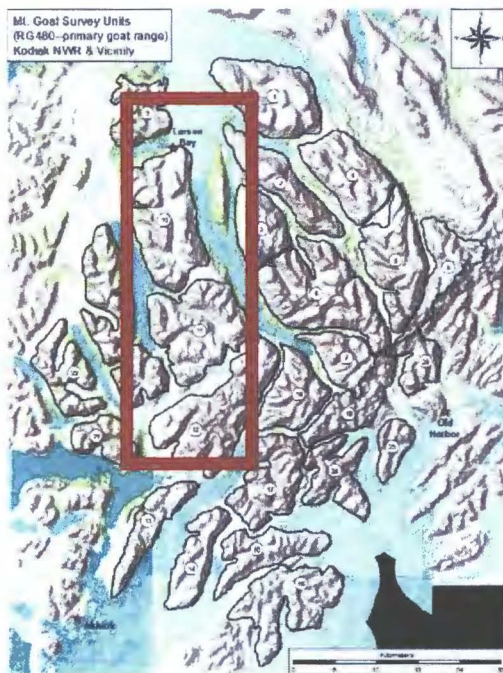
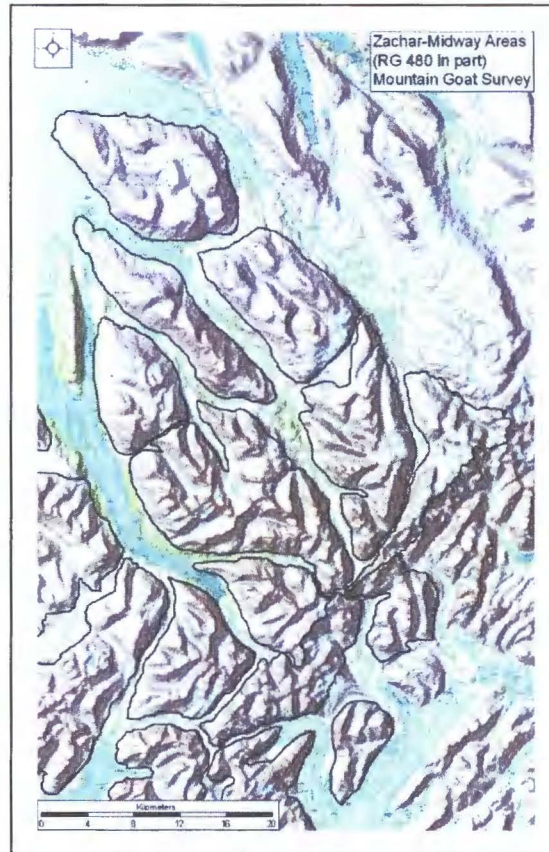


Figure 1. USFWS survey area RG480 in SW Kodiak Island, and 25 survey units within the survey area.

These survey units are grouped into roughly 4 survey sections—Zachar-Midway (survey areas 1-8), East Karluk (9-11), East Red-West Karluk (21-23), Grayback Mt-Barling Bay (12, 17-19, 24-25) (figure 2).



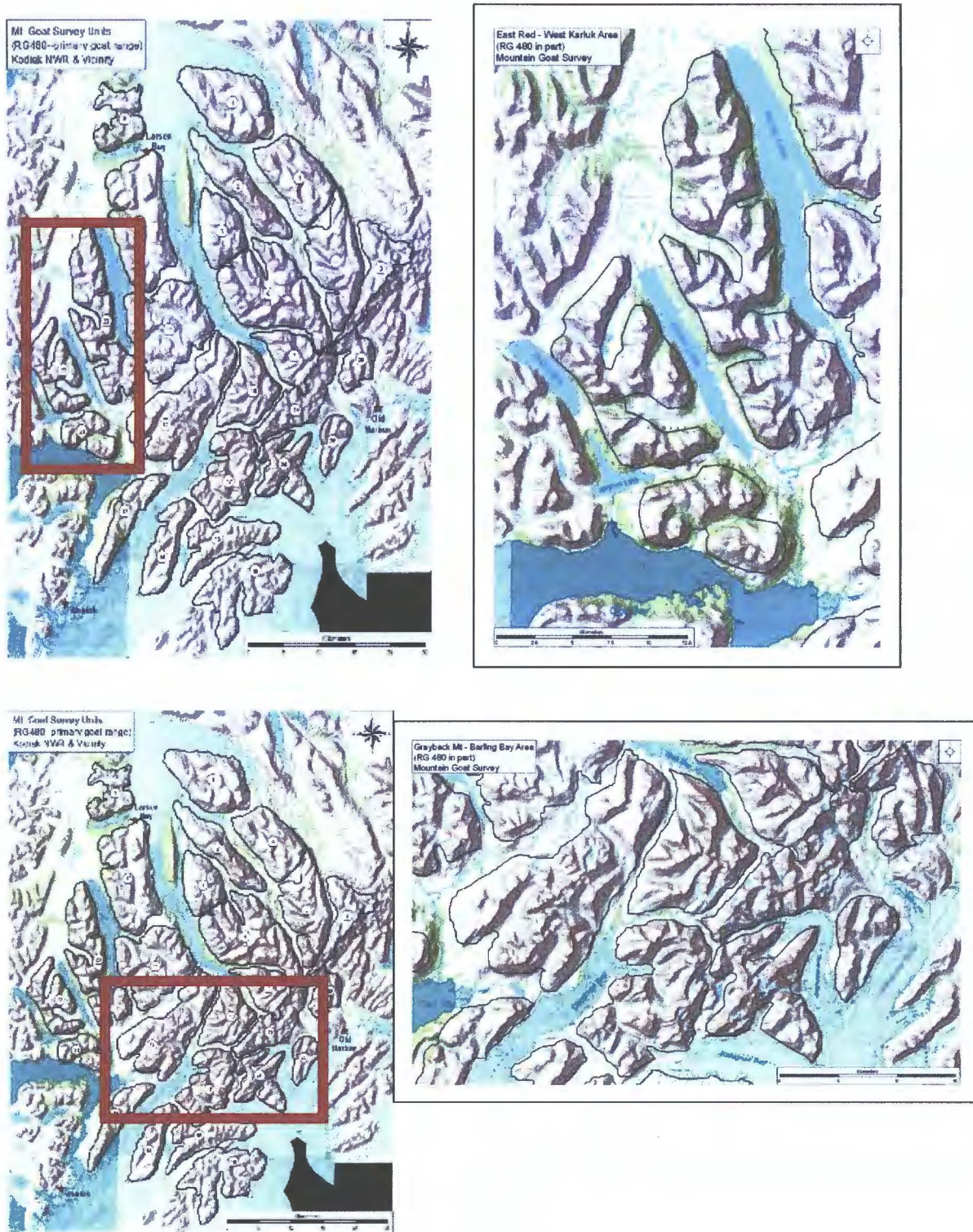


Figure 2. Survey sections

A possible survey example path is shown in figure 3.

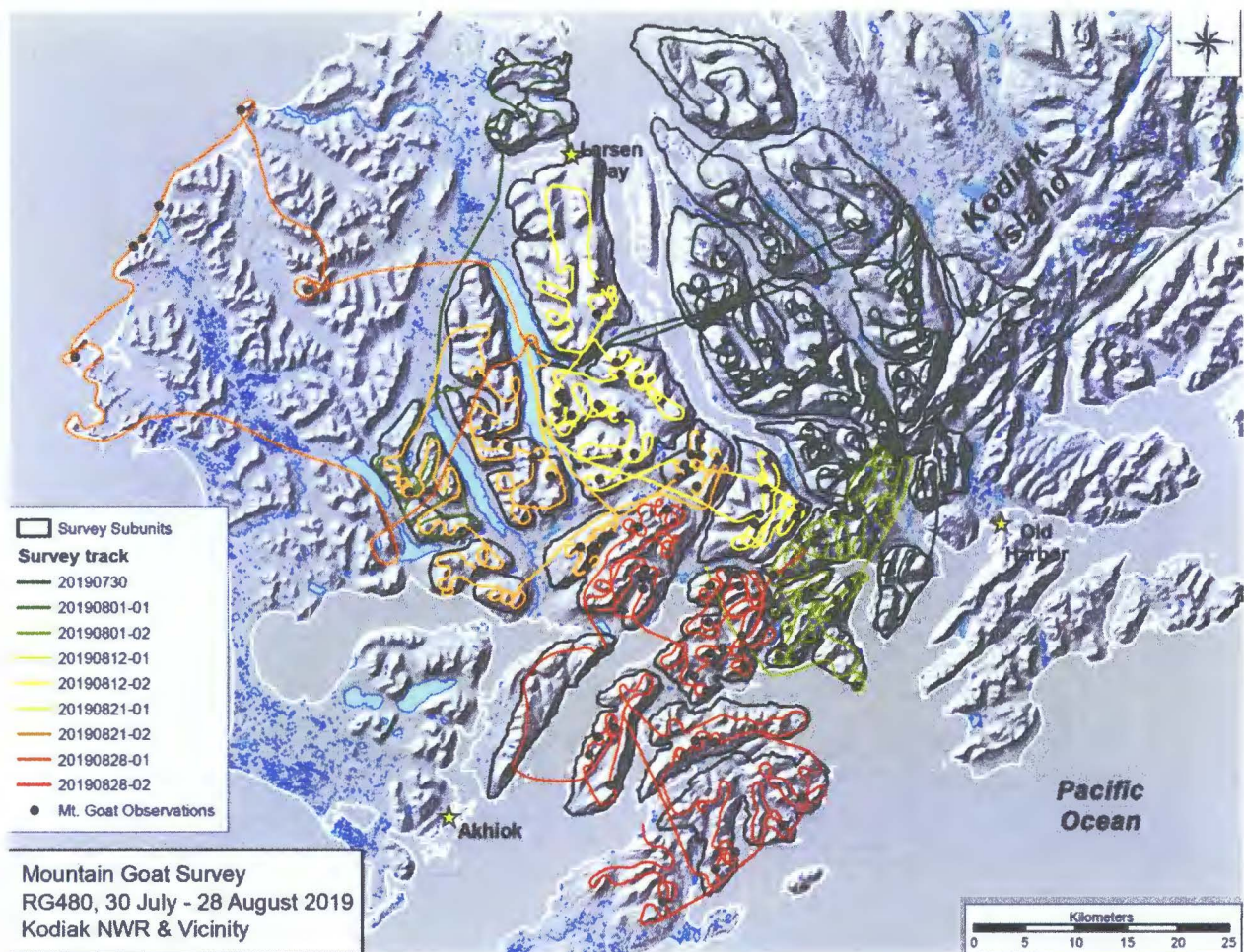


Figure 3. Survey tracks and observations, 2019.

One pilot and one observer from USFWS are used to complete the survey, which includes all 25 survey units.

The survey takes approximately 24 hours, plus ferry time to and from the survey location. It can be broken down into as few as 4 days, but often takes around 9.

Pilot and observer can base out of Camp Island, Karluk Lake to increase efficiency.

Survey efforts should be focused in grass and forage-dominated alpine habitats above shrub-line.

Weather should be overcast to sunny with wind speeds <10 knots and a high ceiling.

The plane should fly around 150 m (500 ft) above ground level while surveying.

Both pilot and observer can count animals.

Upon encountering a mountain goat group, the observer should record the subunit number, GPS waypoint, the group number, time, habitat (alpine, cliff, grass, rock, snow, shrub), total group size, and number of adults and kids.

A group should be separated by >500 m from other goats, and behaviorally responded to observers in a cohesive manner.

If a GPS waypoints directly over the group is not feasible, note and manually adjust the locations of the waypoints during post-processing.

Datasheets (figure 3) should be scanned, data from datasheets should be either entered into excel (bleh!) or entered into the mountain goat database (which is not up to date—only through 2013), waypoints and tracks should be downloaded, and all data should be stored in T:\Biological\Ungulates\MountainGoat\AerialSurveys\data. All data should be provided to ADFG for analyses.

Mountain Goat Aerial Survey

Survey ID (XXXXMMDDSurvey#): _____ Date: _____

Observer: _____ Pilot: _____ Aircraft: _____

Time			
Takeoff _____	Landing _____	Total Flight Time: _____	
Survey start _____	Survey end _____	Total Survey Time: _____	
Total Break Time: _____		Net Survey Time (Total Survey Time - Total Break Time): _____	

Weather			
Survey Start: Cloud cover (%) _____	Precip. <i>none mist rain</i>	Wind: variable 5-10 10-15 >15	Turb: <i>none light moderate</i>
Survey End: Cloud cover (%) _____	Precip. <i>none mist rain</i>	Wind: variable 5-10 10-15 >15	Turb: <i>none light moderate</i>

Notes: _____

GPS Wyp#	Group#	Time	Habitat*	Total	Adults	Kids	Unk.	Notes (including start/end times of breaks)

*Habitat types: A = alpine, C = cliff, G = grass, R = rock/boulder, S = snow, SH = shrub

Figure 3. Datasheet, from T:\Biological\Ungulates\MountainGoat\AerialSurveys\SurveyDataSheet&Maps

Note for future work:

Surveys are only conducted in alpine terrain, so mountain goats occupying lower elevation habitat are not counted. We assume this number is small.

We do not know sightability, and it likely varies across the island, so estimates are not true estimates.

There are no replicates. (so how does ADFG generate estimates?)

Future efforts should be focused on developing a quantitative estimate of goat sightability that can be applied to count data, develop a methodology that would allow for robust confidence interval estimates, and reduce the need to survey the entire island. Conducting island-wide population counts of Kodiak Island mountain goats has become increasingly costly to achieve because of the increasing size and range of this mountain goat population, a method allowing a subset to be surveyed would be more economical.

Appendices:

Example data (T:\Biological\Ungulates\MountainGoat\AerialSurveys\data\2019\
MtGoatSurveyData_RG480_2019.xlsx)

HuntArea	SSU	surveyDate	survey_ID	WP	groupID	time	habitat	utmE	utmN	total	adults	kids	unk	comment
480	1	7/30/2019	2019073001	1	1	7:56	A	461989	6383143	27	20	7	0	
480	2	7/30/2019	2019073001	2	2	8:26	R	463047	6367104	9	9	0	0	
480	2	7/30/2019	2019073001	3	3	8:30	R	460913	6370007	60	40	20	0	
480	3	7/30/2019	2019073001	4	4	8:42	G	456420	6367520	50	36	14	0	
480	3	7/30/2019	2019073001	5	5	8:46	C	455425	6362116	12	5	7	0	

Subunit area from

T:\Biological\Ungulates\MountainGoat\AerialSurveys\data\RG480_SurveySubunitArea.xlsx

Subunit	KM2
1	124
2	85
3	65
4	101
5	112
6	169
7	54
8	66
9	55
10	158
11	144
12	161
13	59
14	63
15	72
16	113
17	90
18	74
19	67
20	41
21	51
22	69
23	146
24	64
25	31